

Package ‘confeR’

November 3, 2025

Version 0.1
Date 2025-11-03
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Title confeR
Description Implements Bayesian conjugate analysis for federated inference as described in Degen et al. (2025) <[doi:XXXXXXXXXXXX](#)>.
License GPL-3
Encoding UTF-8
Imports invgamma, dplyr, tidyr, tibble
Suggests roxygen2
NeedsCompilation no
RoxygenNote 7.3.3
URL <https://gitlab.uzh.ch/crsuzh/bayseq>
BugReports <https://gitlab.uzh.ch/crsuzh/bayseq/issues>

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bayes_local_glm	<i>Local params</i>
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Description

Compute local glm parameters from summary statistics.

Usage

```
bayes_local_glm(bstats, center_identity, family, alpha = 0.05)
```

Arguments

bstats	List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites’.
center_identity	The id of the site to be removed.
family	Family to pass to glm() call, e.g. "gaussian".
alpha	Numeric. Significance level for credible intervals. Default 0.05.

Value

List with beta hat and credible intervals.

Author(s)

Peter Degen

bca_iterate_sites	<i>BCA iterate over local sites</i>
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Description

Pre-processing wrapper for BCA methods.

Usage

```
bca_iterate_sites(
  outcome,
  covariates,
  model,
  family,
  data_split,
  use_local_intercepts,
  center_name = NULL
)
```

Arguments

outcome	Name of outcome y.
covariates	Vector of covariate names.
model	Formula object relating outcome to covariates.
family	Family to pass to glm() call, e.g. "gaussian".
data_split	List, each element containing a dataframe with the data. from a local site.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.
center_name	Character (optional). Name of covariate denoting site identity. Must be supplied if 'use_local_intercepts' is TRUE.

Value

List with transmitted summary statistics for each local site.

Author(s)

Peter Degen

bca_oneshot	<i>BCA one-shot</i>
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Description

Perform one-shot computation of parameter estimates given transmitted summary statistics.

Usage

```
bca_oneshot(
  bstats,
  n_sites,
  use_local_intercepts,
  use_local_variances,
  family,
  covariates,
  epsilon = 1e-10,
  center_name = NULL,
  alpha = 0.05
)
```

Arguments

bstats	List of summary statistics from each local site, e.g. obbject returned by 'bca_iterate_sites'.
n_sites	Number of sites.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.
use_local_variances	Logical. If true, use fixed site-specific residual variances for each local site.

family	Family to pass to glm() call, e.g. "gaussian".
covariates	Vector of covariate names.
epsilon	Numeric. Regularization for prior hyperparameters. Default 1e-10.
center_name	Character (optional). Name of covariate denoting site identity. Must be supplied if 'family' is 'gaussian'.
alpha	Numeric. Significance level for credible intervals. Default 0.05.

Value

List of oneshot parameter estimates

Author(s)

Peter Degen

box_check_all_sites	<i>Box prior predictive tail probabilities for all sites</i>
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Description

Calculate p_Box for all sites.

Usage

```
box_check_all_sites(
  data_split,
  params_oneshot,
  bstats,
  family,
  local_sigma2s,
  covariates,
  n_sites,
  use_local_intercepts,
  center_name
)
```

Arguments

data_split	The data split by local sites.
params_oneshot	Parameters returned by 'bca_oneshot'.
bstats	List of summary statistics from each local site, e.g. obbject returned by 'bca_iterate_sites'.
family	Family to pass to glm() call, e.g. "gaussian".
local_sigma2s	Numeric vector with local residual variances from glm().
covariates	Vector of covariate names.
n_sites	Number of sites.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.
center_name	Character. Name of covariate denoting site identity.

Value

List with pboxes and reduced parameters for all sites.

Author(s)

Peter Degen

get_pred_prob	<i>Box prior predictive tail probability</i>
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Description

Check if estimate from specific site is compatible with posterior from all other centers.

Usage

```
get_pred_prob(
  center_identity,
  bstats,
  local_sigma2s,
  covariates,
  n_sites,
  reduced_params,
  use_local_intercepts
)
```

Arguments

center_identity	The id of the site to be checked
bstats	List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites’.
local_sigma2s	Numeric vector with local residual variances from glm().
covariates	Vector of covariate names.
n_sites	Number of sites.
reduced_params	Reduced posterior parameters returnd by ‘get_reduced_params’.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.

Value

The Box prior predictive tail probability.

Author(s)

Peter Degen

get_reduced_params	<i>Get reduced params</i>
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Description

Use Reverse-Bayes to get reduced parameters obtained by leaving out one local site from the full posterior.

Usage

```
get_reduced_params(center_identity, params_one-shot, bstats, family)
```

Arguments

center_identity	The id of the site to be removed.
params_one-shot	Parameters returned by bca_one-shot.
bstats	List of summary statistics from each local site, e.g. object returned by 'bca_iterate_sites'.
family	Family to pass to glm() call, e.g. "gaussian".

Value

List with reduced parameter estimates.

Author(s)

Peter Degen

tidy_results	<i>BCA tidy results</i>
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Description

Tidy results from one-shot

Usage

```
tidy_results(params_one-shot, use_local_intercepts)
```

Arguments

params_one-shot	Parameters returned by 'bca_one-shot'.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.

Value

Dataframe

Author(s)

Peter Degen

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